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SYSTEM AND METHOD OF DISTRIBUTED COMMUNICATIONS

Abstract

A system and method to facilitate synchronized operation and/or distributed operation of one or more aspects of a secure element of a first device in a communication system, such as an NFC capable communication system of the device, that communicates with a mobile device (e.g., a portable device or smartphone).

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Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure relates to a system and method for communicating with a portable device, and more particularly to communication capabilities for a plurality of devices with the portable device.

BACKGROUND

[0002] Portable devices, also described as mobile devices, have become increasingly prevalent in communications across a wide spectrum of applications. For instance, a portable device may communicate with a real-time locating system, potentially disposed on an object, such as a vehicle or a building, in order to facilitate determining location of the portable device with respect to the object.

[0003] Additionally, or alternatively, the portable device may establish communications to provide, receive, or exchange information in a secure manner. For instance, the portable device may include communication circuitry capable of communicating according to a near field communication (NFC) standard. Conventionally, an NFC enabled device may communicate with the portable device to exchange data, such as transaction information or payment information. The NFC enabled device may be in the form of an NFC reader capable of obtaining information from the portable device via communications according to the NFC standard. The portable device and/or the NFC reader, in conventional systems, may include a secure element (SE) that includes an applet (an application or program with read and/or writable data capabilities) that communicates with another NFC capable device (e.g., another portable device or another NFC reader). The applet may take the form of an electronic wallet that stores a secure payment token to implement a financial transaction in which a physical credit card or debit card associated with the secure payment token is not physically present. In other words, the applet may emulate aspects of the physical card so that the portable device and/or the NFC reader can be used without physical presence of the physical card. Additionally, rather than being a fixed medium, the applet may be adapted to incorporate additional payment tokens or to update software (e.g., to fix a security vulnerability).

SUMMARY

[0004] A system and method are provided to facilitate synchronized operation and/or distributed operation of one or more aspects of a secure element of a first device in a communication system, such as an NFC capable communication system of the device, that communicates with a mobile device (e.g., a portable device or smartphone).

[0005] In one embodiment, a near field communication (NFC) system is provided with a first device and a second device. The first device may include a secure element (SE) stored as secure element data in the first device. The first device may also include a first device controller communicatively coupled to the secure element, and configured to communicate secure element information pertaining to the secure element. The secure element information may include at least one of a) a data packet generated by or provided to the secure element and b) at least a portion of the secure element data of the secure element.

[0006] The second device may be remote from and communicatively coupled to the first device. The second device may include a second device controller configured to communicate the secure element information with respect to the first device controller, and operable to provide a second device secure element interface based on the secure element information. The second device secure element interface may be a remote interface to the secure element of the first device.

[0007] In one embodiment, the second device may include a second device NFC antenna configured to transmit NFC communications to a mobile device and to receive NFC communications from the mobile device. The second device may also include a second device NFC

controller communicatively coupled to the second device NFC antenna, where the second device NFC controller is operable to communicate with the second device secure element interface. The second device NFC controller may be configured to receive, from the mobile device, a mobile device data packet for the second device secure element interface. The remote data packet may correspond to communications transmitted via a communication link between the second device and the mobile device.

[0008] In one embodiment, the secure element information received by the second device controller is a distributed version of a secure element applet of the secure element, where the second device secure element interface is the distributed version of the secure element applet of the secure element.

[0009] In one embodiment, after the mobile device interacts with the distributed version of the secure element applet via the second device secure element interface, the second device controller may provide secure element information to the first device controller and pertaining to a state change with respect to the secure element relating to interactions between the mobile device and the second device secure element interface.

[0010] In one embodiment, the secure element information is at least one of received by the second device controller and transmitted to the first controller, and where the secure element information pertains to the data packet generated by or provided to the secure element, wherein the second device secure element interface relays the mobile device data packet.

[0011] In one embodiment, a method is provided for communicating via near field communication (NFC). The method may include communicating secure element information pertaining to a secure element (SE) of a first device, where the secure element information includes at least one of a) a data packet generated by or provided to the secure element and b) at least a portion of secure element data of the secure element. The method may include communicating, in a second device, the secure element information with the first device, and providing, in the second device that is remote from the first device, a second device secure element interface that is based on the secure element information communicated with the first device.

[0012] In one embodiment, the method may include transmitting NFC communications from the second device to a mobile device, and receiving, in the second device, NFC communications transmitted from the mobile device that includes a mobile device data packet for the second device secure element interface.

[0013] In one embodiment, the method may include obtaining a lock with respect to the secure element of the first device, operating the second device secure element interface as the secure element of the first device, and substantially preventing a third device secure element interface of a third device from operating as the secure element of the first device until after the lock is released.

[0014] In one embodiment, the method may include the second device receiving the secure element information in accordance with a distributed version of a secure element applet of the secure element, and providing the second device secure element interface conducting NFC communications in accordance with the distributed version of the secure element applet of the secure element.

[0015] In one embodiment, the method may include communicating, in the second device, the secure element information with the first device includes relaying the mobile device data packet to the secure element of the first device.

[0016] Before the embodiments of the invention are explained in detail, it is to be understood that the invention is not limited to the details of operation or to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention may be implemented in various other embodiments and of being practiced or being carried out in alternative ways not expressly disclosed herein. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including” and “comprising” and variations thereof is meant to

encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof. Further, enumeration may be used in the description of various embodiments. Unless otherwise expressly stated, the use of enumeration should not be construed as limiting the invention to any specific order or number of components. Nor should the use of enumeration be construed as excluding from the scope of the invention any additional steps or components that might be combined with or into the enumerated steps or components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 shows a representative view of a system in accordance with one embodiment.

[0018] FIG. 2 shows a representative view of the system of FIG. 1 in accordance with one embodiment.

[0019] FIG. 3 shows a representative view of aspects of an object device in accordance with one embodiment.

[0020] FIG. 4 shows a representative view of aspects of a remote device in accordance with one embodiment.

[0021] FIG. 5 shows a method of operation in accordance with one embodiment.

DETAILED DESCRIPTION

[0022] A system and method according to one embodiment are provided to facilitate synchronized operation and/or distributed operation of one or more aspects of a secure element of a first device in a communication system, such as an NFC capable communication system of the first device, that communicates with a mobile device (e.g., a portable device or smartphone). In one embodiment, the first device may share the secure element, or aspects thereof, with one or more remote devices, enabling each of the one or more remote devices to communicate with the mobile device according to the shared portion of the secure element in place of the first device. Additionally, or alternatively, the first device and the one or more remote devices may be operable to communicate information pertaining to the secure element via a communication link to enable a remote device to communicate according to an aspect of the secure element without all or a portion of the aspect being provided on the remote device. For instance, the remote device may relay communications from the mobile device to the first device for processing by the secure element of the first device, and a response from the secure element may be communicated back to the remote device for transmission to the mobile device.

[0023] In one embodiment, the first device may share the secure element, or aspects thereof, with one or more remote devices. Additionally, or alternatively, the first device may share the secure element, or aspects thereof, with a second remote device, and that second device may share the secure element, or aspects thereof, with a third remote device. Communications may occur in reverse as well, with the third remote device sharing information to the second remote device, which then shares information with the first device.

[0024] A system and method according to one embodiment of the present disclosure may allow a mobile device to communicate with a secure element of the first device without communicating directly with the first device. In one embodiment, the mobile device may be capable of communicating wirelessly in accordance with a communication standard that allows for communication over relatively short distances, including, for example, the NFC standard (e.g., less than 10 cm or less than 20 cm). Multiple remote devices may be disposed distal from the first device at a distance that is greater than the effective communication distance of the communication standard (e.g., a distance greater than 10 cm or 20 cm in the case of NFC). Each of the remote devices may be operable to communicate with the mobile device in accordance with the communication standard and in accordance with one or more aspects of the secure element stored

in the first device. This way, the remote devices may increase the effective distance of communications according to the communication standard, and allow the mobile device to communicate according to the communication standard over larger areas. In a practical example, the remote devices may be disposed in a plurality of locations on an object (e.g., a vehicle), and an NFC capable mobile device may communicate with any one of the remote devices according to an aspect of the secure element of the first device. Each of the remote devices may be indistinguishable from each other with respect to the secure element, or an aspect thereof. If the mobile device or the secure element, or both, is configured to pair with the other of the mobile device or the secure element (such as by securely storing information pertaining to the other of the mobile device or the secure element) pairing with each of the remote devices may be avoided. In other words, each of the remote devices may operate in accordance with the pairing so that additional pairing for each remote device can be avoided.

[0025] In one embodiment, the system and method according to one embodiment may facilitate content distribution of a secure element from a first device to one or more remote devices. The secure element may be synchronized among the first device and the one or more remote devices via a communication link. A mobile device may pair with one of the first device and the one or more remote devices, and synchronization or content distribution among the first device and the one or more remote device may enable the mobile device to communicate with any one of the first device and the one or more remote devices as if that device were substantially the same as any other of the first device and the one or more remote devices with respect to the paired communication. The mobile device may not need to pair with each of the first device and the one or more remote devices.

[0026] The communication link between the first device and the one or more remote devices may be a low bandwidth link in one embodiment, such that the bandwidth of the communication link is incapable of effectively relaying communications to and from a remote device and the first device to present the remote device as having the secure element of the first device. Using the communication link, one or more aspects of the secure element of the first device may be communicated to the one or more remote devices (e.g., when possible, such as before communicating with the mobile device) and stored in memory of the one or more remote devices. Communications relating to the secure element, such as communications that occur before communications with the mobile device, pertain to content distribution and/or synchronization among the first device and the one or more remote devices.

[0027] In one embodiment, changes to one or more data items, data, or information associated with a secure element may be synchronized with the first device and the one or more remote devices via a communication link between the first device and the one or more remote devices. The changes may be synchronized periodically or in response to an event, such as in response to an item change, data change, or information change, or any combination thereof. For instance, as discussed herein in further detail, if memory associated with the secure element is modified or updated in one device in response to providing temporary access to an object associated with the device and relative to the mobile device, the changes to memory may be distributed to one or more other devices via the communication link so that any one of the first device and the one or more remote devices may be capable of authenticating access to the object in response to communicating with the mobile device.

[0028] In one embodiment, the secure element or an aspect thereof may be communicated from the first device to one or more remote devices and provided on the one or more remote devices. With multiple copies of the secure element or an aspect thereof being provided in multiple locations, a failure of any one device may be recoverable by replacement of the failed device and providing a copy of the lost data to the new device.

[0029] In one embodiment, memory associated with the secure element may be distributed over a plurality of devices, such as across the first device and a remote device, such that at least a portion

of the memory is stored in one device but not another. This way, in cases where the memory requirements of the secure element are large relative to the available space on any one device, the effective memory of the secure element can be increased.

[0030] In one embodiment, content distribution and/or synchronization of one or more aspects of a secure element may be conducted in a manner that provides distributed operation of one or more applets of the secure element without affecting operation of one or more applets of the secure element or being perceptible to the one or more applets. This way, an applet provided for the system may be generated independent of or without awareness of the content distribution and/or synchronization functionality provided in accordance with one embodiment of the present disclosure. An applet in one embodiment may be configured to facilitate content distribution and/or synchronization in accordance with one or more embodiments described herein. The applet configured in this way may or may not be privileged. Such an applet may be considered a dedicated applet.

[0031] Although one or more embodiments according to the present disclosure may involve synchronization of a secure element, or aspects thereof, across multiple devices, it is to be understood that the present disclosure is not so limited. In one embodiment, the system may include multiple devices each including a secure element or a distributed form thereof. Because there are multiple secure elements in the system, potentially disposed at different locations, the system may provide one or more of reduced latency, additional storage, and additional compute capacity. Such a system may provide an enhanced user experience.

[0032] It is noted that synchronization and/or content distribution in one or more embodiments according to the present disclosure may include one or more versions and/or signatures of summaries of the data in a secure element that is communicated. In this way, the system may determine which secure elements to update. Optionally, the system may determine a course of action for updating the secure elements, or an aspect thereof.

[0033] In one embodiment, memory associated with a secure element may be distributed over multiple devices, such that memory associated with a secure element applet of said secure element is decentralized. This decentralization may be managed by a synchronization protocol that is at least one of distributed, decentralized, and synchronized. Data may be distributed in any manner. For instance, data may be centralized to one secure element and/or data may be copied in multiple secure elements. Additionally, or alternatively, a decentralized protocol may be utilized to synchronize and/or distribute data across a network.

I. System Overview

[0034] A system in accordance with one embodiment is shown in the illustrated embodiment of FIGS. 1 and 2 and generally designated **100**. The system **100** may include one or more system components as outlined herein. A system component may be a user **60**, an object device **50**, or remote device **40** shown in the illustrated embodiments of FIGS. 3 and 4. The system component also may be a mobile device **20** or another electronic component including one or more aspects of the example devices. The underlying components of the system component, as discussed herein, may be configured to operate in conjunction with any one or more system components. In this sense, in one embodiment, there may be several aspects or features common among the mobile device **20**, the remote device **40**, and the object device **50**, as well as other components of the system **100** described herein.

[0035] For instance, one or more features described in connection with the object device **50** depicted in FIG. 3 may form part of a mobile device **20** or the remote device **40**, or any combination thereof. Conversely, one or more features described in conjunction with the mobile device **20** or the remote device **40** may form part of the object device **50**. Additionally, or alternatively, it is to be understood that any feature described in conjunction with any of the object device **50**, mobile device **20**, remote device **40**, and any system component, may be absent in one or more embodiments. In one embodiment, the object device **50** or a component of a control system

of an object **10** may include the same or similar components described in conjunction with a system component, and may form a component disposed on the object **10**, such as a vehicle or a building.

[0036] An object device **50** in the form of a system component may be communicatively coupled to one or more systems of the object **10**, collectively forming an object control system to control operation of the object **10**. Information may be communicated among system components of the object control system, such as by being transmitted and received between two or more components of the object **10**.

[0037] As mentioned herein, the object **10** including the object device **50** may include communication capabilities. The object **10** may include one or more communication networks **150**, wired or wireless, that facilitate such communication, such as a wired bus shown in the illustrated embodiment of FIG. **1**. The communication network **150** may also enable one or more system components, internal or external to the object control system, to communicate with the object control system. For instance, the communication network **150** may facilitate communication between the object control system (including one more system components as described herein) and the object device **50**. Such a communication network **150** may be a CAN bus and is shown as a vehicle bus in the illustrated embodiment of FIG. **2**. Additionally, or alternatively, the object control system may facilitate communication directly or indirectly among system components. For instance, the object control system in the illustrated embodiment of FIG. **1** is configured to communicate and enable direct communication between the object device **50** and an object control **12**, embodied to include an engine control module (ECM).

[0038] In one embodiment, the object **10** may include a telematics control unit (TCU), which is not shown. For instance, the TCU may be connected to the object control system via the communication network **150** or another type of communication link, such as an SPI link. In another embodiment, the TCU may be combined with the object control system, such as being part of the object control **12**. The TCU may be absent in one embodiment and data that would have been provided by the TCU may be tunneled through the mobile device **20** (e.g., via BTLE). “Tunneled” may be defined as a traditional tunnel-like running TCP/IP over BTLE; however, the present disclosure is not so limited. The tunnel may be defined as a configuration that enables relevant data to be communicated to the object control system or other system components via commands/responses.

[0039] In one embodiment, the TCU may include a cellular modem or other long range WAN radio (Lora, Sigfox, etc.).

[0040] In one embodiment, as described above, the TCU is not a required part of the system; for instance, all functionality of the TCU and the system it communicates with may be performed locally (e.g., not in the cloud).

[0041] In the illustrated embodiment of FIGS. **1** and **2**, the object **10** is provided with an object device **50** and a plurality of remote devices **40** disposed at positions relative to the object **10**. For instance, in the context of the vehicle shown in FIG. **2**, the remote devices **40** are disposed inside or near the vehicle door **14**, inside or near the vehicle rearview mirror **15**, or a variety of other locations of the vehicle. Example locations are also described in U.S. Pat. No. 10,356,550 to Smith et al., entitled METHOD AND SYSTEM FOR ESTABLISHING MICROLOCATION ZONES, filed Dec. 14, 2017, issued Jul. 16, 2019, and U.S. Pat. No. 10,362,461 to Stitt et al., entitled SYSTEM AND METHOD FOR MICROLOCATION SENSOR COMMUNICATION, filed Dec. 22, 2017, issued Jul. 23, 2019—the disclosures of which are hereby incorporated by reference in their entirety.

[0042] The remote devices **40** in one embodiment may be sensors or monitor devices capable of detecting communications with respect to the mobile device **20** and another system component, such as the object device **50** or another remote device **40**. In one embodiment, the remote devices **40** may communicate sensed information (e.g., signal strength, time of flight, angle of arrival)

pertaining to communications detected with respect to the mobile device **20**. For instance, the remote device **40** may communicate sensed information via a communication link **130** to another device, such as the object device **50**, connected to the communication link **130**. The object device **50** may be operable to determine a location of the mobile device **20** based on the sensed information. Examples of such a determination are also described in U.S. Pat. No. 10,356,550 to Smith et al., entitled METHOD AND SYSTEM FOR ESTABLISHING MICROLOCATION ZONES, filed Dec. 14, 2017, issued Jul. 16, 2019, and U.S. Pat. No. 10,362,461 to Stitt et al., entitled SYSTEM AND METHOD FOR MICROLOCATION SENSOR COMMUNICATION, filed Dec. 22, 2017, issued Jul. 23, 2019—the disclosures of which are hereby incorporated by reference in their entirety.

[0043] Based on the determined location of the mobile device **20** relative to the object **10**, the object device **50** may transmit a command or instruction to the object control **12** to enable a capability of the object **10**, such as to mobilize the object **10** in the case of a vehicle, or to enable access to a location related to the object **10**. The object device **50** in one embodiment may include a locator capable of receiving sensor information pertaining to wireless communications with the mobile device **20** (e.g., Bluetooth Low Energy (BTLE) and/or Ultra Wide Band (UWB) communications), including one or more signal characteristics of the communications, such as signal strength (e.g., RSSI), Angle of Arrival, and Time of Flight. In one embodiment, as described herein, the object device **50** may include a secure element controller **220** capable of facilitating communications with a secure element **230** of the object device **50**. The secure element **230** may vary from application to application. For example, the secure element **230** provides secure storage and processing capability with any of the following configurations: a) a separate module/IC from a host (application) module, with its own storage and execution software (operating system and applets), b) a separate module/IC from a host (application) module, with its own storage, but with its execution software (operating system and applets and/or application software) in the host module, c) an isolated integrated module of a host (application) module, with its own storage and execution software (operating system and applets), d) an isolated integrated module of a host (application) module, with its own storage, but with its execution software (operating system and applets and/or application software) in the host module, e) an integrated component of a host (application) module software, with its storage isolated from other host (application) components via software or hardware mechanisms (with its execution software [operating system and applets and/or application software] operating as an isolated or integrated component of the host module (application) software). In other words, the secure element **230** may be a separate module (SE), an integrated module (HSM), or part of the software with secured storage, or a different configuration depending on the application.

[0044] The secure element **230** may include a container [e.g., a processor] and memory, and the memory may store the OS (e.g., JCOP or something else) of the secure element **230**. Running within the OS are Applets (applications) and data (application data). The secure element **230** may include both RAM and ROM. The secure element **230** may include an attached host processor; however, the secure element **230** may not require such a separate host processor, and thus the secure element **230** may be the only processor of the device. For instance, a first device and a second device may be connected via some network (e.g., CAN, ethernet) and the processing of that network may occur within an applet in the secure element **230** or as part of the operating system software of the secure element **230**.

[0045] In one embodiment, a secure element **230** may derive data as part of communications with a portable device that needs to be distributed. For instance, an SE applet **232** may compute a key, and this key may be shared/distributed to other SEs in the system **100**.

[0046] In the illustrated embodiment, the secure element **230** includes memory separate from the memory **212**. However, the present disclosure is not so limited. For instance, the memory of the secure element **230** may correspond to the memory **212** of the object device **50**, such that the secure

element **230** shares memory **212** of the object device **50**.

[0047] As described herein, the object device **50** and a remote device **40** may include NFC capabilities. The NFC may be coupled indirectly or directed to the secure element **230** of the object device **50**. The NFC may be coupled indirectly or directed to a component of the remote device **40** pertaining to the secure element **230** (e.g., a distributed version of the secure element **230**).

[0048] Communications with the secure element **230** may be conducted by one or more of the remote devices **40** in a distributed or shared manner as described herein, such that secure element information is transmitted from the object device **50** to the remote device **40** via the communication link **130**. The communication link **130** may be a wired connection or wireless connection, or a combination thereof. As an example, the communication link **130** may be established via a BTLE connection, or the communication link **130** may be established via a CAN bus.

[0049] The mobile device **20** in the illustrated embodiment may be capable of communicating via a communication link **140** with one or more of the remote devices **40** and the object device **50** according to a communication standard. The communication link **140** in the illustrated embodiment is an NFC communication link, but the present disclosure is not so limited.

[0050] The communication link **140** in one embodiment is not limited to a single type of communication link or establishment of a single communication link at a given time between the mobile device **20** and a system component of the object **10**. For instance, the mobile device **20** may establish a first communication link **140** with a first remote device **40** according to the NFC standard, and may simultaneously establish a second communication link **140** with the object device **50** according to the BTLE standard. As another example, the mobile device **20** may establish a first communication link **140** with a first remote device **40** at one time according to the NFC standard, and establish a second communication link **140** with a second remote device **40** at another time according to the NFC standard or according to the BTLE standard.

[0051] In the illustrated embodiment, the communication link **140** is shown optionally present, in phantom lines, between the mobile device **20** and each of the object device **50** and the plurality of remote devices **40**. It is noted that, as discussed herein, one or more of these communication links **140** may be established at any given time, and that multiple communication links **140** may be established simultaneously or concurrently (with the same device or separate devices).

[0052] As discussed herein, the object device **50** may include a secure element controller **220** capable of facilitating communication with respect to a secure element **230** in a distributed or shared manner across more than one device. Such communication may be achieved via one or more of the communication links **140**, such that an effective range for communications in accordance with an aspect of the secure element **230** may be increased. Additionally, or alternatively, the communication via one or more of the communication links **140** via a plurality of devices (e.g., two or more of the object device **50** and the plurality of remote devices **40**) may enhance the robustness of available communication between the mobile device **20** and an aspect of the secure element **230**.

II. Object Device

[0053] The object device **50** in accordance with one embodiment is shown in FIG. 3. The object device **50** may include a control system **200**. The control system **200** of the object device may include one or more processors **210** that execute one or more applications (software and/or firmware), one or more communication interfaces **214**, and a secure element controller **220** operable to facilitate communication with respect to a secure element **230** of the object device **50**. In one embodiment, the secure element controller **220** may be an NFC controller operably coupled to the communication interface **214**. The object device **50**, as described herein, may include memory **212** (e.g., RAM and/or ROM), which may be internal or external to the control system **200**—although it is shown external to the control system **200** in the illustrated embodiment. It is noted that the present disclosure is not limited to use of NFC in conjunction with the secure element **230**. For instance, instead of an NFC controller, the secure element **230** may operate in conjunction with any wired or wireless technology, including a BLE or UWB controller, or an application processor

not connected to radios.

[0054] In the illustrated embodiment, the communication interface **214** may be optionally coupled to one or more antennas **217** (e.g., an NFC antenna). As described herein, although the object device **50** includes the secure element **230** for NFC communications, the object device **50** may not necessarily be capable of communicating wirelessly with an NFC capable device—e.g., the mobile device **20**. Rather, in one embodiment, the object device **50** may communicate secure element information with one or more remote devices **40**, which may be capable of communicating wirelessly with an NFC capable device in accordance with the NFC standard. Alternatively, the object device **50** may include an NFC antenna, and the secure element controller **220** may communicate with the mobile device **20** in accordance with the NFC standard.

[0055] The control system **200** of the object device **50** may include an operating system that controls access to lower level electronics of the control system **200** via the communication interface **214**, which may operate as an interface between one or more components of the control system **200** or between one or more components of the control system **200** and one or more components of the object device **50** external to the control system **200**. For instance, the operating system may facilitate transmission and reception of communications to an object device communicator **215**, which may be configured to facilitate communication with one or more remote devices **40** disposed on the object **10**. The object device communicator **215** may be operable to establish the communication link **130** between the object device **50** and one or more of the remote devices **40**. It is to be understood that the object device communicator **215** is not necessarily an auxiliary communicator or secondary communicator, and may form the primary communication circuitry for establishing external communications with respect to the object device **50**. In one embodiment, the object device communicator **215** may facilitate establishing the communication link **130** as well as exchanging communications via the communication network **150**. For instance, the communication link **130** and the communication network **150** may share a physical medium or transmission of communications signals, and the object device communicator **215** may be operable to generate and process such communications signals.

[0056] In one embodiment, the secure element controller **220** may be operable to communicate directly with one or more components coupled to the communication interface **214**, such as an antenna **217** or the object device communicator **215**, or both. A secure element controller **220**, for instance, may bypass an operating system of the control system **200** to allow direct interface between a secure element applet **232** of the secure element **230** and a component coupled to the communication interface **214**, such as an antenna **217** or the object device communicator **215**, or both.

[0057] The object device communicator **215** may provide any type of communication link, including any of the types of communication links describe herein, including wired or wireless. The communication interface **214** may facilitate communication in conjunction with the object device communicator **215** that is external or internal, or both. As an example, the communication interface **214** in conjunction with the object device communicator **215** may provide a wireless communication link with another system component in the form of the remote device **40**, such as wireless communications according to the BTLE standard, or an external server component (e.g., the cloud) via a Wi-Fi Ethernet communication link. In another example, the communication interface **214** in conjunction with the object device communicator **215** may be configured to communicate with an object component (e.g., a vehicle component) via a wired link such as a CAN-based wired network that facilitates communication between a plurality of devices. The control system **200** in one embodiment may include a display and/or input interface coupled to the communication interface **214** for communicating information to and/or receiving information from the user **60**.

[0058] The object device **50**, and one embodiment, may be configured to communicate with one or more auxiliary devices (not shown) other than a system component or a user **60**. The auxiliary

device may be configured differently from a system component, such as by not including a processor and instead including at least one direct connection and/or a communication interface for transmission or receipt, or both, of information with the object device **50**. For instance, the auxiliary device may be a solenoid that accepts an input from the object device **50**, or the auxiliary device may be a sensor (e.g., a proximity sensor) that provides analog/and or digital feedback to the object device **50**.

[0059] The secure element **230**, in the illustrated embodiment, is shown separate from memory **212**; however, one or more aspects of the secure element **230** may be stored in memory **212**, which may be tamper-resistant memory for security sensitive applications, such as the secure element **230** and one or more secure element applets **232** provided in the secure element **230**. The secure element **230** in the illustrated embodiment is the secure element controller **220**, which may facilitate communications with the secure element **230** in accordance with one or more embodiments herein, including communications with one or more remote devices **40** to enable distributed and/or shared use of the secure elements **230** by devices other than the object device **50**. The secure element controller **220**, in one embodiment, may include an NFC communication controller that handles communications to and from the secure element **230** in conjunction with NFC related communications with the mobile device **20** and the object device **50** or a remote device **40**.

[0060] The secure element **230** may be operable to generate data packets and to receive data packets from a secure element controller **220**. The secure element controller **220** may receive and/or transmit such data packets as secure element information or part of secure element information communicated via the communication interface **214**.

[0061] The secure element **230**, as described herein, may include one or more secure element applets **232**, also described as an applet **232**. An applet **232** may be an application or type of software configured to enable a particular task or set of instructions. The applet **232** may include an interface with one or more readable and/or writable memory portions to facilitate performance of the task or set of instructions. The memory portions may or may not be provided in the memory **212**. The applet **232** may form a closed or managed software application separate from user accessible aspects of the object device **50**. In other words, the applet **232** may perform the task or set of instructions separate from other aspects of the control system **200** such that the modifications to the control system **200** are potentially incapable of modifying instructions or accessing memory associated with the applet **232** without doing so through an application program interface of the applet **232**. This way, an applet **232** can be considered a controlled set of instructions to enable a particular task in a secure manner. Updates or modifications to the applet **232** may be conducted via an application program interface of the applet **232** and/or the secure element **230**. For instance, the secure element **230** may allow removal or installation of an applet **232** within memory space of the secure element **230**. Likewise, the secure element **230** may allow modifications or updates to the applet **232** if authorization to do so is established.

[0062] Examples of applets **232** include an electronic wallet capable of facilitating financial services or payment processing without a physical card (e.g., without a credit card or debit card). Additional examples of applets **232** include a credential store capable of facilitating authentication and/or authorization with respect to one or more actions, commands, or instructions of the object **10**.

[0063] An applet **232** may be operable to communicate with an external device, such as a point-of-sale terminal. Alternatively, the applet **232** may form the point-of-sale terminal for communicating with a corresponding applet of a mobile device **20**.

[0064] The secure element **230**, as discussed herein, may be configured to store information in memory **212**. Such information may be stored by the applet **232** or another aspect of the secure element **230**. Example types of information include credentials, authorization information, and authentication information used for facilitating secure communications requesting an action in a

secure manner. For instance, the information may be authenticating a mobile device **20** and determining the mobile device **20** (e.g., or a user **60** of the mobile device **20**) is authorized with respect to a command or action pertaining to the object **10**. In one embodiment, the applet **232** may facilitate authenticating and authorizing a request from the mobile device **20** to unlock a vehicle door **14**. Additionally, or alternatively, the vehicle door **14** may be unlocked in conjunction with a location determination with respect to the mobile device **20**. For instance, the object device **50** may determine that the location of the mobile device **20** is within a zone relative to the vehicle door **14** that is associated with unlocking or enabling the vehicle door **14** to be unlocked, and the secure element **230** may communicate with the mobile device **20** in order to determine whether the mobile device **20** is authenticated and/or authorized to enable the vehicle door **14** to be unlocked.

III Remote Device

[0065] The remote device **40** in accordance with one embodiment is shown in FIG. **4**. The remote device **40** may include a control system **300**, which may be similar to the control system **200** of the object device **50** in several respects. For instance, the control system **300** of the remote device **40** may include one or more processors **310** that execute one or more applications (software and/or firmware), one or more communication interfaces **314**, and a secure element controller **320**.

[0066] The secure element controller **320** of the remote device **40** may facilitate communication with respect to the secure element **230** of the object device **50**. In one embodiment, the secure element controller **320** may be an NFC controller operably coupled to the communication interface **314**. The remote device **40**, as described herein, may include memory **312** (e.g., RAM and/or ROM), which may be internal or external to the control system **300**—although it is shown external to the control system **300** in the illustrated embodiment.

[0067] In the illustrated embodiment, the communication interface **314** may be coupled to one or more antennas **317** (e.g., an NFC antenna). The remote device **40**, via the communication interface **314** and the antenna **317**, may establish a communication link **130** with the mobile device **20**. The communication link **130** may be established according to the NFC standard.

[0068] The remote device **40** in the illustrated embodiment may include a remote device communicator **315** similar to the object device communicator **215**, and capable of facilitating establishment of the communication link **130** with the object device **50** in conjunction with the communication interface **314** of the remote device **40**. The secure element controller **320** of the remote device **40** may directly interact with the communication interface **314**, similar to the communication interface **214** and secure element controller **220** of the object device **50**.

[0069] In the illustrated embodiment, the secure element controller **320** may include a secure element interface **340**, which, as described herein, may form a remote or distributed interface with respect to the secure element **230**.

[0070] In one embodiment, the secure element interface **340** of the secure element controller **320** may be operable as a pass-through or relay for communications established between the mobile device **20** and the remote device **40**. Information communicated to and from the secure element **340** of the secure element controller **320** and the secure element controller **220** of the object device **50** may be considered secure element information. In other words, communications from the mobile device **20** directed to the secure element controller **320** may be transmitted to the secure element controller **220** via the communication link **130** or processing by the secure element **230**.

Communications from the secure element **230** of the object device **50** may be transmitted to the secure element controller **320** of the remote device **40** and passed on to the mobile device **20** via the communication link **140**. The communications in this example may be in accordance with the NFC standard; but it is to be understood the present disclosure is not so limited. The communications may be established in accordance with an additional or alternative standard.

[0071] The secure element interface **340** in one embodiment, by operating as a pass-through for communications directed to the secure element **230** of the object device **50**, may enable the remote device **40** to operably communicate with the mobile device **20** as if the remote device **40** included a

local version of the secure element **230** of the object device **50** despite the secure element **230** being stored in the object device **50**.

[0072] In one embodiment, as described herein, the system **100** may include a plurality of remote devices **40**, each including a secure element interface **340**. The mobile device **20** may establish a communication link **140** with any one of the remote devices **40** and communicate with the secure element **230** of the object device **50** via the secure element interface **340** of the remote device **40**. The mobile device **20** may be unaware that the secure element **230** is physically located at a remote location relative to the remote device **40**. The communication link **140** between the mobile device **20** and the remote device **40** may be in accordance with the NFC standard, which may allow for communications over a short range, such as less than 20 or 10 cm. The remote device **40** may be disposed at a distance much greater than the short range relative to the object device **50**, allowing the mobile device **20** to communicate with the secure element **230** over a range greater than the short range associated with the communication link **140**.

[0073] In one embodiment, the distance over which the communication link **140** may be extended to communicate with the secure element **230** of the object device **50** may be limited by the communication capabilities of the communication link **130** between the remote device **40** and the object device **50**.

[0074] It is to be understood that the communication link **130** between the remote device **40** and the object device **50** may include sufficient bandwidth to support pass-through or relaying of communications directed to and generated from the secure element **230** of the object device **50** with respect to communications between the mobile device **20** and the remote device **40** via the communication link **140**. For instance, the bandwidth of the communication link **130** may be greater than or equal to the bandwidth of the communication link **140** in order to avoid performance degradation with respect to the communication link **140** and communications with the secure element **230** of the object device **50**.

[0075] In the case where a plurality of remote devices **40** are present, the system **100** may be capable of facilitating communications with the secure element **230** of the object device **50** over an effective range or distance much greater than would be otherwise possible with communication capabilities limited to direct communication between the object device **50** and the mobile device **20**. For instance, in the case of a vehicle as the object **10** and the communication link **140** being an NFC type of communication link, the remote devices **40** may be disposed at various locations on the vehicle to enable NFC communications with the secure element **230** over an effective range much greater than NFC supports. The mobile device **20** may communicate with the secure element **230** via a remote device **40** at several locations on the vehicle.

[0076] With a remote device **40** configured in accordance with one embodiment, the mobile device **20** may communicate with the remote device **40** without needing to specifically pair with the remote device **40** for use with a secure element or secure element applet of the mobile device **20**. The mobile device **20** may pair with the secure element **230** of the object device **50**, and communicate with the secure element **230** via any one of the remote devices **40**.

[0077] In one embodiment, the secure element interface **340** of the remote device **40** may be operable to obtain a distributed version of the secure element **230** (or an aspect thereof) of the object device **50** via the communication link **140** with the object device **50**. For instance, the secure element interface **340** may obtain a copy of the secure element **230** or a copy of one or more of the applets **232** of the secure element **230**. The distributed version of the secure element **230** or an aspect thereof may be stored in memory **312** of the remote device **40**, and is shown in phantom lines as a secure element distributed version **350** and a secure element distributed applet **352**.

[0078] In the illustrated embodiment, the secure element distributed version **350** may be substantially similar to the secure element **230** of the object device **50**. The secure element interface **340** may be operable to synchronize or push changes in the secure element distributed version **350** to the object device **50** so that the secure element **230** is up-to-date with respect to such changes.

Conversely, the secure element controller **220** may be operable to push changes in the secure element **230** to the secure element controller **320** of the remote device **40** so that the secure element distributed version **350** is up-to-date.

[0079] In one embodiment, the secure element interface **340** may communicate with the secure element controller **220** to obtain a lock with respect to the secure element **230**. The secure element interface **340** may communicate with the mobile device **20** in conjunction with the secure element distributed version **350**, and any changes thereto may be pushed or synchronized to the secure element **230** in an atomic or serialized manner. The lock may prevent operations on the secure element **230** that relate to operations being conducted by the secure element interface **340**. It is to be understood that a lock may not be used in cases where memory locations associated with the secure element **230** and the secure element distributed version **350** are not changed in response to communications between the mobile device **20** of the remote device **40** via the communication link **140**. In this way, the secure element interface **340** may conduct atomic operations with respect to the secure element **230** while interfacing with the secure element distributed version **350**.

[0080] With a distributed form of the secure element **230**, the remote device **40** may communicate with the mobile device **20** in accordance with the secure element **230** without directly communicating with the secure element **230** of the object device **50**. As discussed herein, changes pertaining to the secure element **230** that are generated by the remote device **40** may be transmitted as secure element information by the secure element controller **322** and the secure element controller **220** of the object device **50**. This way, the remote device **40** may facilitate synchronizing such changes with the secure element **230** of the object device **50**.

[0081] In one embodiment, the bandwidth of the communication link **140** between the object device **50** and the remote device **40** may be insufficient to facilitate pass-through or relaying of information between the secure element controller **320** and the secure element controller **220** of the object device **50** in real-time to facilitate communications with the mobile device **20**. For instance, the bandwidth of an NFC communication link between the mobile device **20** and the secure element controller **320** may be greater than the available bandwidth for the communication link **140** between the remote device **40** and the object device **50**. To avoid such bandwidth limitations, the secure element distributed version **350** may be utilized by the secure element interface **340** and present the mobile device **20** with an interface substantially identical to the secure element **230** or an aspect thereof in a manner that is substantially undetectable to the mobile device **20**. The mobile device **20** in this configuration may not need to pair with a secure element of each remote device **40**, and instead may be presented with a substantially unified interface in the form of the secure element **230** via the plurality of remote devices **40** and the object device **50**.

[0082] In one embodiment, with distributed versions of the secure element **230** being provided on one or more remote devices **40**, and with the one or more remote devices **40** being separate from the object device **50**, the system **100** may provide a failure recovery mode in case of a memory failure or device failure (e.g., failure of an object device **50** or a remote device **40**). For instance, if the object device **50** fails or a number or memory becomes corrupt, a new object device **50** may be installed to replace the defective object device **50**. The secure element controller **220** and/or the secure element controller **320** may communicate with each other to synchronize with an up-to-date secure element **230**.

[0083] In one embodiment, the secure element interface **340** may store portions of the secure elements **230** of the object device **50** in the memory **312** of the remote device **40**. The portions stored in the memory **312** of the remote device **40** may be absent from the memory of another device, such as the memory **312** of another remote device and/or the memory **212** of the object device **50**. As a result, memory associated with the secure element **230** may be distributed across more than one device in the system **100**. The secure element controller **320** and the secure element **230** may be operable to obtain memory not present in a respective device via the communication link **130** if such memory is or becomes required in accordance with communications with a mobile

device **20** via the communication link **140**. In one embodiment, distributing memory associated with the secure element **230** across multiple devices in the system **100** may enable the system **100** to increase an effective amount of memory associated with the secure element **230**.

IV. Method of Operation

[0084] A method of operation in accordance with one embodiment according to the present disclosure is depicted in FIG. 5 and generally designated **1000**. The method may include communicating secure element information with respect to a secure element **230** stored in a first device, such as an object device **50**. Step **1002**. The secure element information may correspond to a secure applet **232** of the secure element **230**, or the secure element information may correspond to information directed to or transmitted from the secure applet **232** of the secure element **230**.

[0085] A second device, such as a remote device **40**, may receive or transmit, or both, the secure element information with respect to the first device. Step **1004**. For instance, as described herein, the second device may receive a distributed version of the secure applet **232** of the secure element, or the second device may receive information generated from the secure applet **232** stored in the first device. The second device may provide a second device secure element interface based on the secure element information received from the first device. Step **1006**.

[0086] The second device secure element interface, in one embodiment, may correspond to a distributed version of the secure applet **232** of the secure elements **230**. The second device may obtain a lock with respect to the secure applet **232** in order to prevent read operations with respect to the secure applet **232** stored in the first device. While the lock is present, read operations may be permitted for the secure applet **232** stored in the first device, including read operations with respect to distributed versions of the secure applet **232** provided on devices other than the first and second devices.

[0087] The second device secure element interface may facilitate NFC communications with a mobile device separate from and movable with respect to the first and second devices. Steps **1008**, **1010**. The second device secure element interface may also facilitate transmitting information to the first device that is based on the NFC communications received from the mobile device. Step **1012**.

[0088] In one embodiment, with the second device secure element interface providing a distributed version of the secure applet **232**, NFC communications with the mobile device may be substantially identical to NFC communications that would otherwise occur with respect to the secure applet **232** stored in the first device. In this way, the second device secure element interface may enable NFC communications in accordance with the secure applet **232**, despite the second device being remote from the first device at a distance for which NFC communications may not be possible between the first device and the mobile device. This arrangement may effectively increase an effective range of NFC communications between a system of devices, including first and second devices, and the mobile device.

[0089] The second device secure element interface, in one embodiment, may facilitate relaying information between the secure element **230** stored in the first device and a mobile device. For instance, NFC communications received from the mobile device and directed to the secure elements **230** may be relayed from the second device to the first device via a communication link separate from the NFC communications between the mobile device and the second device. Information from the secure element **230** of the first device may be received by the second device, and the second device secure element interface may generate NFC communications based on this information received from the first device.

[0090] For purposes of disclosure, the method **1000** is described in conjunction with first and second devices. It is to be understood, however, that additional devices may interact with the first device and a manner similar to the second device. A method of operating a third device in this manner is depicted in the illustrated embodiment in phantom lines, with steps **1014**, **1016**, **1018**, **1020**, **1022** corresponding respectively to steps **1004**, **1006**, **1008**, **1010**, and **1012**.

[0091] Directional terms, such as vertical, horizontal, top, bottom, upper, lower, inner, inwardly, outer and outwardly, are used to assist in describing the invention based on the orientation of the embodiments shown in the illustrations. The use of directional terms should not be interpreted to limit the invention to any specific orientation(s).

[0092] The above description is that of current embodiments of the invention. Various alterations and changes can be made without departing from the spirit and broader aspects of the invention as defined in the appended claims, which are to be interpreted in accordance with the principles of patent law including the doctrine of equivalents. This disclosure is presented for illustrative purposes and should not be interpreted as an exhaustive description of all embodiments of the invention or to limit the scope of the claims to the specific elements illustrated or described in connection with these embodiments. For example, and without limitation, any individual element(s) of the described invention may be replaced by alternative elements that provide substantially similar functionality or otherwise provide adequate operation. This includes, for example, presently known alternative elements, such as those that might be currently known to one skilled in the art, and alternative elements that may be developed in the future, such as those that one skilled in the art might, upon development, recognize as an alternative. Further, the disclosed embodiments include a plurality of features that are described in concert and that might cooperatively provide a collection of benefits. The present invention is not limited to only those embodiments that include all of these features or that provide all of the stated benefits, except to the extent otherwise expressly set forth in the issued claims. Any reference to claim elements in the singular, for example, using the articles a, an, the or said, is not to be construed as limiting the element to the singular. Any reference to claim elements as at least one of X, Y and Z is meant to include any one of X, Y or Z individually, and any combination of X, Y and Z, for example, X, Y, Z; X, Y; X, Z; and Y, Z.

Claims

1. A communication system comprising: a first device including: a secure element (SE) including secure element data; a first device controller communicatively coupled to said secure element, said first device controller configured to communicate secure element information pertaining to said secure element, said secure element information including at least one of a) a data packet generated by or provided to said secure element and b) at least a portion of said secure element data of said secure element; a second device being remote from said first device, said second device communicatively coupled to said first device, said second device including: a second device controller configured to communicate said secure element information with said first device controller, said second device controller operable to provide a second device secure element interface based on said secure element information, said second device secure element interface being a remote interface to said secure element of said first device; a second device antenna configured to transmit communications to a mobile device and to receive communications from the mobile device; said second device controller configured to receive, from the mobile device, a mobile device data packet for said second device secure element interface; and wherein said secure element information is at least one of received by said second device controller and transmitted to said first device controller, wherein said secure element information pertains to said data packet generated by or provided to said secure element, wherein said second device secure element interface relays the mobile device data packet to said secure element via said first and second device controllers to operate as said secure element.

2. The system of claim 1 comprising: a third device being remote from said first and second devices, said third device communicatively coupled to said first device, said third device including: a third device controller configured to communicate said secure element information with said first device controller, said third device controller operable to provide a third device secure element

interface based on said secure element information, said third device secure element interface being a remote interface to said secure element of said first device; a third device antenna configured to transmit and receive communications from the mobile device; and said third device controller configured to receive, from the mobile device, a mobile device data packet for said third device secure element interface.

3. The system of claim 1 wherein said second device controller is operable to obtain a lock with respect to said secure element of said first device, wherein said second device secure element interface is operable as said secure element of said first device, and wherein a third device secure element interface is substantially prevented via said lock from operating as said secure element of said first device until after said second device controller releases said lock.

4. The system of claim 3 wherein said lock is obtained with respect to a secure element applet of said secure element, such that another secure element applet of said secure element remains unlocked relative to said lock.

5. The system of claim 1 wherein said secure element information received by said second device controller is a distributed version of a secure element applet of said secure element, wherein said second device secure element interface is said distributed version of said secure element applet of said secure element.

6. The system of claim 5 wherein, after or while the mobile device interacts with said distributed version of said secure element applet via said second device secure element interface, said second device controller provides secure element information, to said first device controller, pertaining to a state change with respect to said secure element relating to interactions between the mobile device and said second device secure element interface.

7. The system of claim 1 wherein said first device includes: a first device antenna configured to transmit communications to the mobile device and to receive communications from the mobile device; and said first device controller configured to receive, from the mobile device, a mobile device data packet for said secure element.

8. The system of claim 1 wherein said first device controller is operable to provide a distributed interface to content of said secure element.

9. The system of claim 1 wherein said first device controller and said second device controller are operable to establish a secure communication connection, wherein said secure element information is transmitted via said secure communication connection.

10. The system of claim 1 wherein said second device includes a distributed version of said secure element, and wherein said distributed version of said secure element forms said second device secure element interface, and wherein said distributed version provides a copy of said secure element of said first device.

11. The system of claim 1 wherein memory associated with said secure element is distributed over multiple devices, such that memory associated with a secure element applet of said secure element is decentralized.

12. A method of communicating, the method comprising: communicating secure element information pertaining to a secure element (SE) of a first device, wherein the secure element information includes at least one of a) a data packet generated by or provided to the secure element and b) at least a portion of secure element data of the secure element; communicating, in a second device, the secure element information with the first device; providing, in the second device that is remote from the first device, a second device secure element interface that is based on the secure element information communicated with the first device; transmitting communications between the second device and a mobile device; receiving, in the second device, communications transmitted from the mobile device that includes information for the second device secure element interface; and wherein communicating, in the second device, the secure element information with the first device includes relaying the information from the mobile device to the secure element of the first device.

- 13.** The method of claim 12 comprising: communicating, in a third device, the secure element information with the first device; providing, in the third device that is remote from the first device, a third device secure element interface that is based on the secure element information communicated with the first device; transmitting communications between the third device and the mobile device; and receiving, in the third device, communications transmitted from the mobile device that includes information for the third device secure element interface.
- 14.** The method of claim 12 comprising: obtaining a lock with respect to the secure element of the first device; operating the second device secure element interface as the secure element of the first device; and substantially preventing a third device secure element interface of a third device from operating as the secure element of the first device until after the lock is released.
- 15.** The method of claim 12 wherein: communicating the secure element information in the second device with the first device includes the second device receiving the secure element information in accordance with a distributed version of a secure element applet of the secure element; and conducting, via the second device secure element interface, communications in accordance with the distributed version of the secure element applet of the secure element.
- 16.** The method of claim 15 comprising, after or while the mobile device interacts with the distributed version of the secure element applet via the second device secure element interface, providing secure element information, to the first device, that pertains to a state change with respect to the secure element, wherein the state change is based on interactions between the mobile device and the second device secure element interface.
- 17.** The method of claim 15 comprising distributing memory associated with the secure element applet of the secure element to multiple devices to provide a decentralized form of the secure element applet.
- 18.** The method of claim 12 wherein communicating, in the second device, the secure element information with the first device includes relaying communications from the secure element of the first device to the mobile device.
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